

- Business Requirements Document (BRD)**
- Project Title:** Data Integration and Aggregation for Customer and Order Data
- Project Overview:**

The objective of this project is to design and implement a data integration and aggregation process for customer and order data stored in CSV files. The process involves loading the data into delta tables, performing data cleansing, creating an SCD type 2 table for order summary data, and aggregating customer spend data.

- Functional Requirements:**

1. **Data Ingestion**

- Read source CSV data from the volume and load it into the delta tables "customer" and "order".
- Customer data is located at: ``Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata``
- Order data is located at: ``Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata``

2. **Data Schema**

- The schema of the "customer" table is: ``CustId`, `Name`, `EmailId`, `Region``
- The schema of the "order" table is: ``OrderId`, `ItemName`, `PricePerUnit`, `Qty`, `Date`, `CustId``

3. **Data Cleansing**

- Remove null and duplicate records from both "customer" and "order" tables.

4. **Order Summary Table Creation**

- Create the "ordersummary" table if it does not exist in the catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard".
- The schema of the "ordersummary" table is: ``CustId`, `Name`, `EmailId`, `Region`, `OrderId`, `ItemName`, `PricePerUnit`, `Qty`, `Date``

5. **SCD Type 2 Table Implementation**

- Join the "customer" and "order" data using the "CustId" field and load the data into an SCD type 2 table "ordersummary".
- Include logic to update the SCD type 2 "ordersummary" table whenever there is a change in the "customer" table.
- Old records should be made inactive, and new records should be made active.
- Correspondingly, the ``StartDate`` and ``EndDate`` should be updated.

6. **Customer Aggregate Spend Table Creation**

- Create a new table "customeraggregatespend" with columns "Name", "TotalAmount", and "Date" in the catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard", if it does not exist.

7. **Data Aggregation**

- Aggregate the "TotalAmount" column from the "ordersummary" table, grouping by the "Name" and "Date" columns.
- Load the aggregated data into the "customeraggregatespend" table.
- **Non-Functional Requirements:**

1. **Data Quality:** The data integration and aggregation process should ensure data accuracy, completeness, and consistency.
2. **Performance:** The process should be optimized for performance to handle large volumes of data.
3. **Scalability:** The solution should be designed to scale with increasing data volumes and complexity.

- **Assumptions and Dependencies:**

1. The input CSV files are in the correct format and location.
2. The delta tables "customer" and "order" are created with the correct schema.
3. The catalog "gen_ai_poc_databrickscoe" and schema "sdic_wizard" exist in the target data storage system.

- **Success Criteria:**

1. The data is successfully loaded into the delta tables "customer" and "order".
2. The "ordersummary" table is created and populated with the joined data.
3. The SCD type 2 logic is implemented correctly, and the "ordersummary" table is updated accordingly.
4. The "customeraggregatespend" table is created and populated with the aggregated data.

- **Testing and Validation:**

1. Verify the data quality and accuracy after data cleansing and aggregation.
2. Test the SCD type 2 logic to ensure correct updates to the "ordersummary" table.
3. Validate the aggregated data in the "customeraggregatespend" table.

By following this BRD, the project team should be able to design and implement a reliable and efficient data integration and aggregation process for customer and order data.